



ABBOTSLEIGH

Student's Name: _____

Student Number:

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Teacher's Name: _____

2023
HIGHER SCHOOL CERTIFICATE
Assessment 4
Trial Examination

Mathematics Extension 2

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen.
- **NESA approved** calculators may be used.
- **NESA approved** reference sheet is provided.
- All necessary working should be shown in every question.
- Make sure your HSC candidate Number is on the front cover of each booklet.
- Start a new booklet for Each Question.
- Answer the Multiple Choice questions on the answer sheet provided.
- If you do not attempt a whole question, you must still hand in the Writing Booklet, with the words '**NOT ATTEMPTED**' written clearly on the front cover.

Total marks – 100

- Attempt Sections I and II.
- All questions are of equal value.

Section I Pages 3 - 9

10 marks

- Attempt Questions 1–10.
- Allow about 15 minutes for this section.

Section II Pages 9 - 19

90 marks

- Attempt Questions 11– 16.
- Allow about 2 hrs and 45 minutes for this section.

Outcomes to be assessed:

Mathematics Extension 2

HSC :

A student

- | | |
|----------------|---|
| MEX12-1 | understands and uses different representations of numbers and functions to model, prove results and find solutions to problems in a variety of contexts |
| MEX12-2 | chooses appropriate strategies to construct arguments and proofs in both practical and abstract settings |
| MEX12-3 | uses vectors to model and solve problems in two and three dimensions |
| MEX12-4 | uses the relationship between algebraic and geometric representations of complex numbers and complex number techniques to prove results, model and solve problems |
| MEX12-5 | applies techniques of integration to structured and unstructured problems |
| MEX12-6 | uses mechanics to model and solve practical problems |
| MEX12-7 | applies various mathematical techniques and concepts to model and solve structured, unstructured and multi-step problems |
| MEX12-8 | communicates and justifies abstract ideas and relationships using appropriate language, notation and logical argument |

SECTION I

10 marks

Attempt Questions 1 – 10

Use the multiple-choice answer sheet

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9

(A) ☐ (B) ☒ (C) ☐ (D) ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

(A) ☒ (B) ☒ (C) ☐ (D) ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows.

(A) ☒ (B) ☒ (C) ☐ (D) ☐

correct →

1 Consider the following statements:

1. $z \bar{z} = |z|^2$
2. $\bar{z}^{-1} = \frac{z}{|z|^2}$

Which of the following is correct about the above statements:

- A. Only statement 1 is true.
- B. Only statement 2 is true.
- C. Both statement 1 and 2 are true.
- D. Neither statement 1 or 2 are true.

- 2 What is the modulus of $\frac{4+2i}{1-2i}$
- A. $2\sqrt{5}$
- B. 4
- C. 3
- D. 2
- 3 Ms Rennie states that if you cross the Pacific Highway using the overbridge, you won't get hurt. The contrapositive of this statement is:
- A. If you don't cross the Pacific Highway using the overbridge, you will get hurt.
- B. If you get hurt, then you didn't cross the Pacific Highway using the overbridge.
- C. Crossing the Pacific Highway without using the overbridge means you will get hurt was not stated by Ms Rennie.
- D. If you cross the Pacific Highway using the overbridge, you could get hurt.
- 4 The indefinite integral $\int x^3(x^4-1)^2 dx$ is:
- A. $\frac{1}{12}(x^4-1)^2 + C$
- B. $\frac{1}{4}(x^4-1)^3 + C$
- C. $x(x^4-1) + C$
- D. $\frac{1}{12}(x^4-1)^3 + C$

- 5 A particle moving in a straight line has its' velocity, v , given by $v = k(a - x)$, where a is a constant and x is the particle's displacement from the point O .

If the particle is initially at O , which of the following is an expression for x ?

A. $x = a(1 + e^{-kt})$

B. $x = a(1 + e^{kt})$

C. $x = a(1 - e^{-kt})$

D. $x = a(1 - e^{kt})$

- 6 If $\hat{a}$, $\hat{b}$ and $\hat{c}$ are mutually perpendicular unit vectors, then $|\hat{a} + \hat{b} + \hat{c}|$ is:

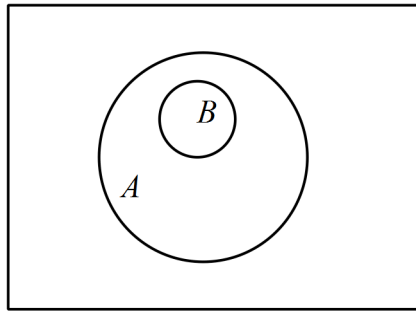
A. 1

B. $\sqrt{2}$

C. $\sqrt{3}$

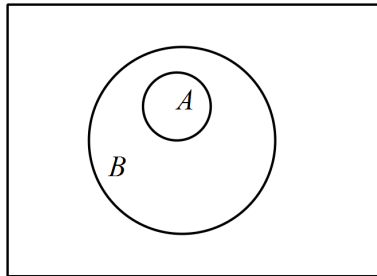
D. 2

7 The Venn diagram below shows $B \Rightarrow A$:

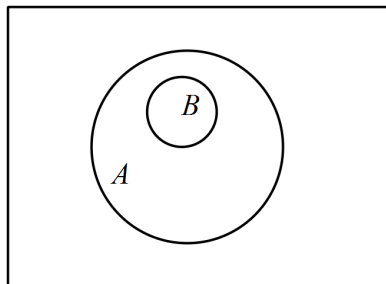


Which diagram below represents the $\neg A \Rightarrow \neg B$?

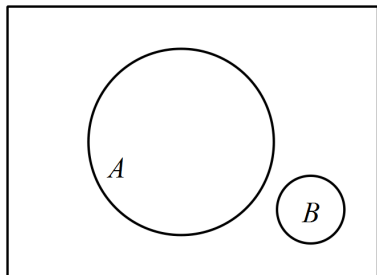
A.



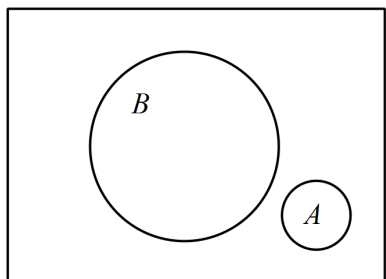
B.



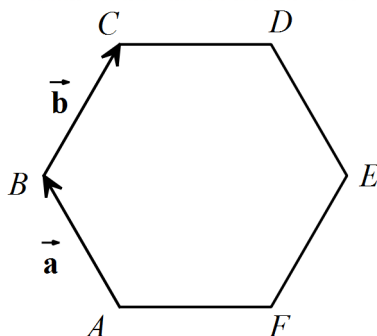
C.



D.



- 8 If $\vec{a}$ and $\vec{b}$ are the vectors forming consecutive sides of a regular hexagon $ABCDEF$, as shown, then the vector representing the side CD is:



- A. $\vec{a} + \vec{b}$
- B. $\vec{a} - \vec{b}$
- C. $\vec{b} - \vec{a}$
- D. $-(\vec{a} + \vec{b})$
- 9 Which of the following integrals evaluates to the largest value?

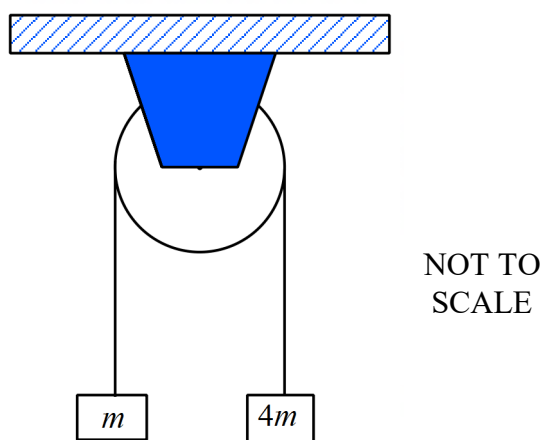
A. $\int_{-\pi}^{\pi} x \sin x \, dx$

B. $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos(x^3) \, dx$

C. $\int_{-\pi}^{\pi} e^{-x^2} \, dx$

D. $\int_{-1}^1 \tan^{-1} x^3 \, dx$

- 10 Below is a diagram showing two objects with masses of m and $4m$ kg on either end of light inextensible strings that pass through a smooth pulley. Both objects are released from rest simultaneously.



Let g be the acceleration due to gravity. After 4 seconds, which of the following is true?

- A. The heavier object has a speed of $\frac{3g}{5} \text{ ms}^{-1}$.
- B. The heavier object has travelled $\frac{24g}{5}$ metres.
- C. The heavier object has an acceleration of $\frac{3g}{4} \text{ ms}^{-2}$.
- D. The heavier object has stopped moving as the lighter object has hit the pulley.

End of Section I

SECTION II

Total Marks – 90

Attempt Questions 11 - 16

All questions are of equal value

Answer each question in a **SEPARATE** writing booklet. Extra writing booklets are available.

In Questions 11-16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet. **Marks**

- (a) If $z = 3 + 2i$ and $w = 2 - i$, find, the following, writing your answer in the form $a + ib$, $a, b \in \mathbb{R}$:

(i) $z \bar{w}$ **2**

(ii) $\frac{z}{i w}$ **2**

(b) Find $\int \frac{x+34}{(x-6)(x+2)} dx$ **3**

(c) Prove that $\sqrt{3}$ is irrational. **2**

(d) Find $\int \frac{dx}{2x^2 + 3x + 4}$ **3**

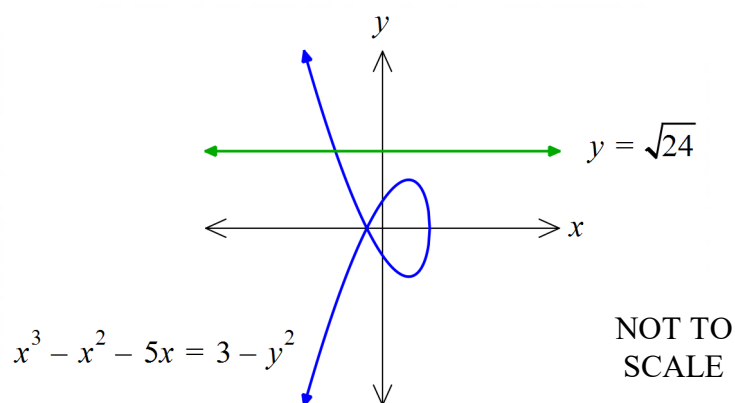
(e) Given $2 + i$ is a zero of $P(x) = x^3 - x^2 - 7x + 15$, find the other zeroes. **3**

End of Question 11

(a) Given $\int_2^3 f(x) dx = \sqrt{7}$, find $\int_1^2 \frac{1}{x^2} f\left(1 + \frac{2}{x}\right) dx$.

3

- (b) The equation $x^3 - x^2 - 5x = 3 - y^2$ implicitly defines the curve shown below.
The line $y = \sqrt{24}$ intersects this curve as shown.



It can be shown that the equation $x^3 - x^2 - 5x + 21 = 0$ will determine the intersection between the line $y = \sqrt{24}$ and the implicitly defined curve.

- (i) Explain, with reference to the graph above, why we know that there is only one real and two complex solutions to this cubic equation.
- (ii) Determine the two exact complex solutions to the equation $x^3 - x^2 - 5x + 21 = 0$.

2

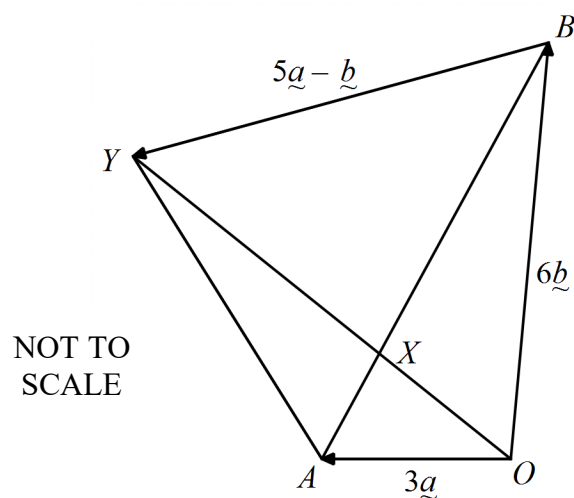
2

Question 12 continues on page 11

Question 12 (continued)

Marks

- (c) The diagram below shows quadrilateral $OAYB$ with $\overrightarrow{OA} = 3\mathbf{a}$ and $\overrightarrow{OB} = 6\mathbf{b}$.



- (i) Express $\overrightarrow{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$. 1
- (ii) X is the point on AB such that $AX : XB = 1 : 2$ and $\overrightarrow{BY} = 5\mathbf{a} - \mathbf{b}$. 3
 Prove that $\overrightarrow{OX} = \frac{2}{5}\overrightarrow{OY}$.
- (d) Let $a, b, c \in \mathbb{R}$.
- (i) Prove the inequality:
- $$a^2 + b^2 \geq 2ab \quad \text{1}$$
- (ii) Hence or otherwise, prove:
- $$a^2 + b^2 + c^2 \geq ab + bc + ca \quad \text{1}$$
- (iii) Hence show:
- $$3(ab + bc + ca) \leq (a + b + c)^2 \leq 3(a^2 + b^2 + c^2) \quad \text{2}$$

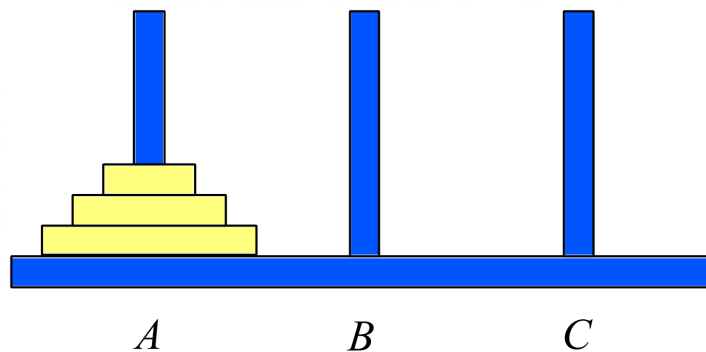
End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

Marks

- (a) The Tower of Hanoi involves moving a set of circular disks from one peg, A , to another peg, C , one at a time, using a “helper tower”, B , and no larger disk can ever be above a smaller disk. **3**

Prove by induction that the number of steps required to move n disks from one peg to another is $2^n - 1$.

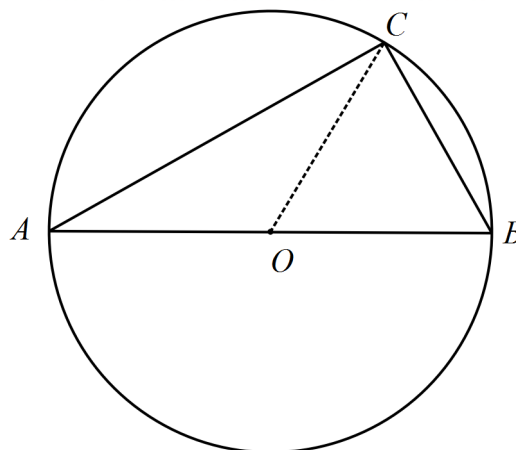


- (b) The acceleration of a particle moving along the x -axis is given by $a = x(2 - 3x) \text{ ms}^{-2}$. Initially, $x = 0 \text{ m}$, $v = 2\sqrt{2} \text{ ms}^{-1}$.

(i) Find an expression for v^2 as a function of x . **2**

(ii) Determine the values that x can take. **2**

- (c) Using vectors, prove that the angle at the circumference of a semi-circle from the ends of the diameter is a right angle. **3**



NOT TO
SCALE

Question 13 continues on page 13

Question 13 (continued)**Marks**

- (d) A particle is moving along a straight line with simple harmonic motion, centre at O , and period $\frac{\pi}{3}$ seconds. **5**

When the particle is 0.48 m from O , its' speed is 2.16 ms^{-1} .

Calculate the total time in one complete oscillation that the particle has speed less than 2.88 ms^{-1}

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

Marks

(a) Prove by induction that $a^4 - 1$ is divisible by 16 for all odd integers, a . **3**

(b) Given $I_n = \int_0^{\sqrt{3}} (3 - x^2)^n dx$.

(i) Show that $I_n = \frac{6n}{2n+1} I_{n-1}$, $n \geq 1$. **3**

(ii) Hence evaluate I_3 . **1**

(c) On the Argand diagram provided as an insert, sketch and label:

(i) $A = \{z : z\bar{z} = 4, z \in C\}$. **1**

(ii) $B = \left\{z : |z| = \left|z - 2cis\frac{\pi}{4}\right|, z \in C\right\}$, labelling the axis intercepts. **2**

(iii) On the diagram, shade the region defined by: **1**

$$\{z : z\bar{z} \leq 4, z \in C\} \cap \{z : \operatorname{Re}(z) + \operatorname{Im}(z) \geq \sqrt{2}, z \in C\}.$$

(iv) Find the area of the shaded region in part (iii). **2**

(d) Prove that, for $x > 0$, $\ln x \leq x - 1$. **2**

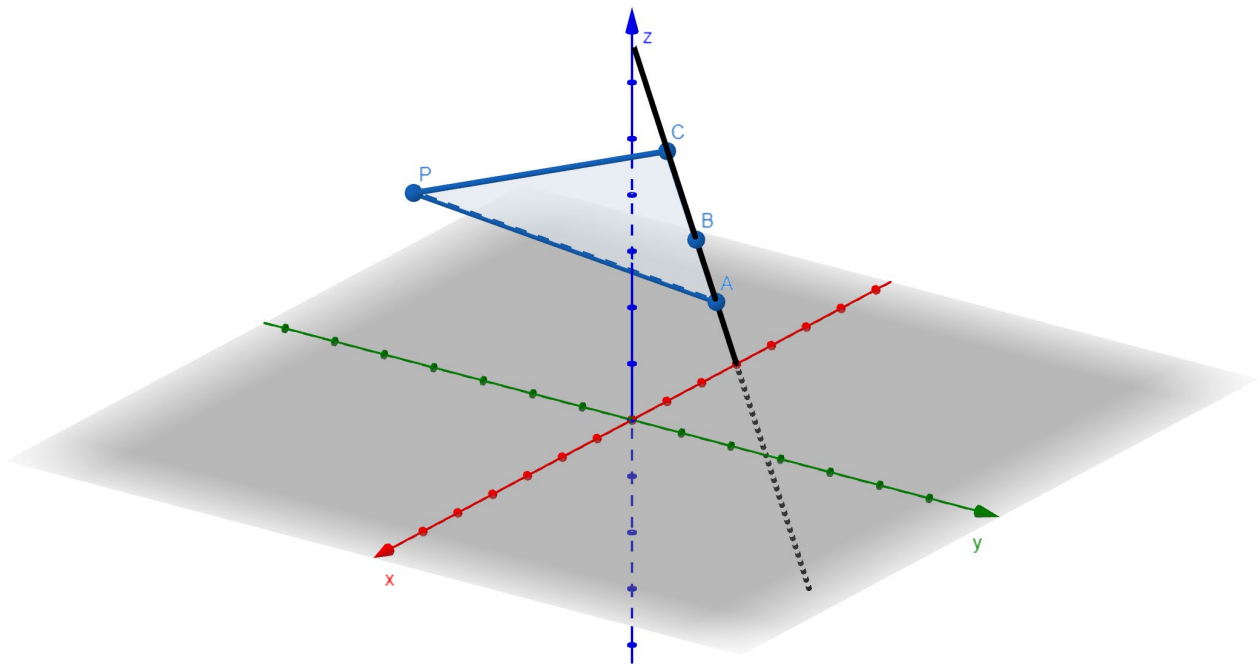
End of Question 14

- (a) Consider the line, l , with vector equation $\underline{r}(\lambda) = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} + \lambda \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$ and the points

$A = (-1, 1, 2)$ and $B = (1, 2, 4)$ lie on the line. Let $\underline{b} = \overrightarrow{AB}$, so

$$\underline{b} = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} \text{ (DO NOT prove this).}$$

Let P be the point $(2, -3, 4)$.



- (i) Find the projection of $\overrightarrow{AP}$ onto the line l , and hence the perpendicular distance from P to the line, l . **3**
- (ii) Hence find the coordinates of the point, C on the line l such that the area of $\triangle APC$ is 15 square units. **2**

Question 15 continues on page 16

(b) The infinite series C and S are defined by:

$$C = \cos \theta + \frac{1}{2} \cos 5\theta + \frac{1}{4} \cos 9\theta + \dots$$

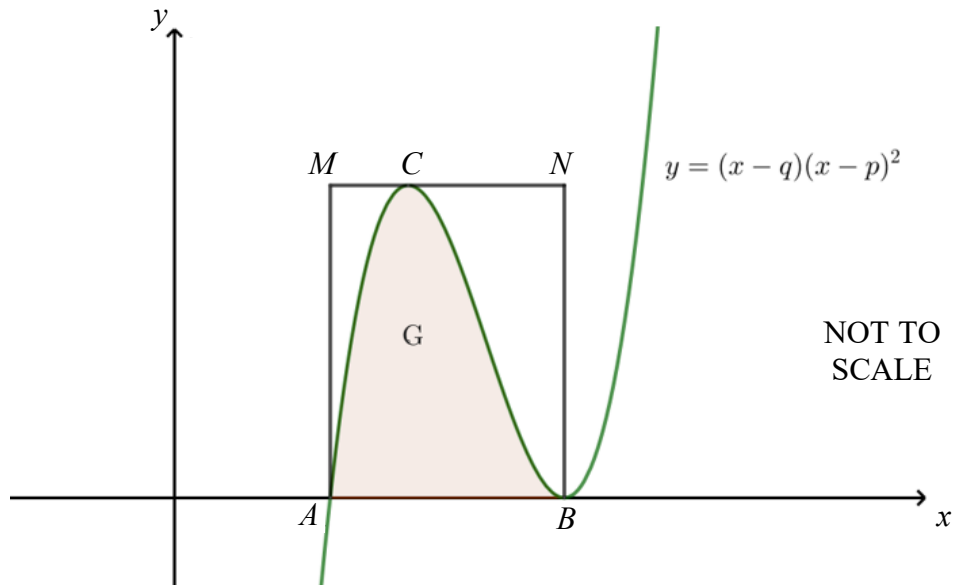
$$S = \sin \theta + \frac{1}{2} \sin 5\theta + \frac{1}{4} \sin 9\theta + \dots$$

(i) Show that $C + iS = \frac{2e^{i\theta}}{2 - e^{4i\theta}}$, given the two series are convergent. 2

(ii) Hence show $S = \frac{4 \sin \theta + 2 \sin 3\theta}{5 - 4 \cos 4\theta}$. 3

Question 15 continues on page 17

- (c) The diagram below shows the curve with equation $y = (x - q)(x - p)^2$, where p and q are positive constants.



The curve meets the x -axis at the points A and B .

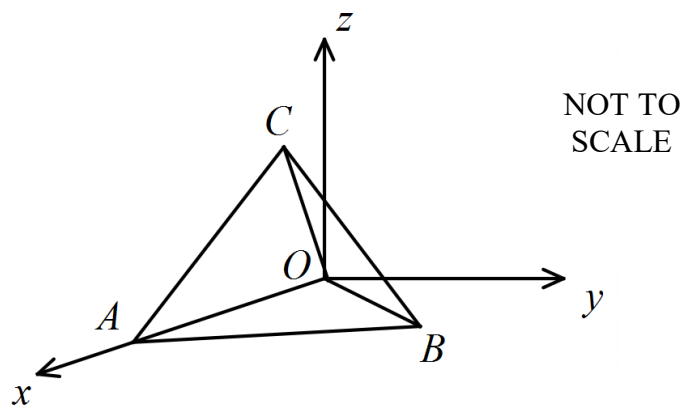
The shaded region, G , is bounded by the curve and the x -axis.

- (i) Show that the area of the shaded region is $\frac{1}{12}(p - q)^4$ 2
- (ii) The rectangle $AMNB$ passes through the local maximum at C . 3
- Show that the area of $AMNB$ is $\frac{16}{9}$ times as big as the region G , regardless of the values of p and q .

End of Question 15

- (a) The faces of the tetrahedron $OABC$ comprise equilateral triangles of side length one unit. Its' base, OAB lies on the xy -plane. Two of the vertices are O and $A(1,0,0)$. The vertex C is above the xy -plane. **4**

Show that the coordinates of C are $\left(\frac{1}{2}, \frac{\sqrt{3}}{6}, \frac{\sqrt{6}}{3}\right)$.



- (b) Let $a, b, c \in \mathbb{R}^+$.

(i) Prove the inequality: $\frac{a}{b} + \frac{b}{a} \geq 2$ **1**

(ii) Hence or otherwise prove: $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}$ **3**

Question 16 continues on page 18

Question 16 (continued).**Marks**

- (c) An object of mass 1 kg is dropped from a hot air balloon. The object is acted on by gravity and experiences a resistive force of kv Newtons, where k is a constant of proportionality.

- (i) Taking down as the positive direction and using $F = mg - kv$ as the equation for the resultant force, show that the velocity, v , of the object t seconds after being dropped is given by: **2**

$$v = \frac{g}{k}(1 - e^{-kt})$$

- (ii) Two seconds after the first object is released, a second object of mass 1 kg is projected downward with an initial velocity of $u \text{ ms}^{-1}$. It is also acted upon by gravity and a resistive force of kv Newtons. Show that the velocity, V , of this second object t seconds after the first object is released is given by: **2**

$$V = \frac{g}{k} - e^{-k(t-2)} \left(\frac{g - ku}{k} \right)$$

- (iii) If the second object is projected at its' terminal velocity $\left(ie: u = \frac{g}{k} \right)$, find an expression for the distance fallen by the second object, y_2 , t seconds after the first object is released. **2**

- (iv) It can be shown that the vertical distance, y_1 , travelled by the first object is given by: **1**

$$y_1 = \frac{g}{k} \left(t + \frac{1}{k} (e^{-kt} - 1) \right) \quad \text{DO NOT prove this.}$$

Show that the time that the two objects will collide is given by:

$$t = -\frac{1}{k} \ln(1 - 2k).$$

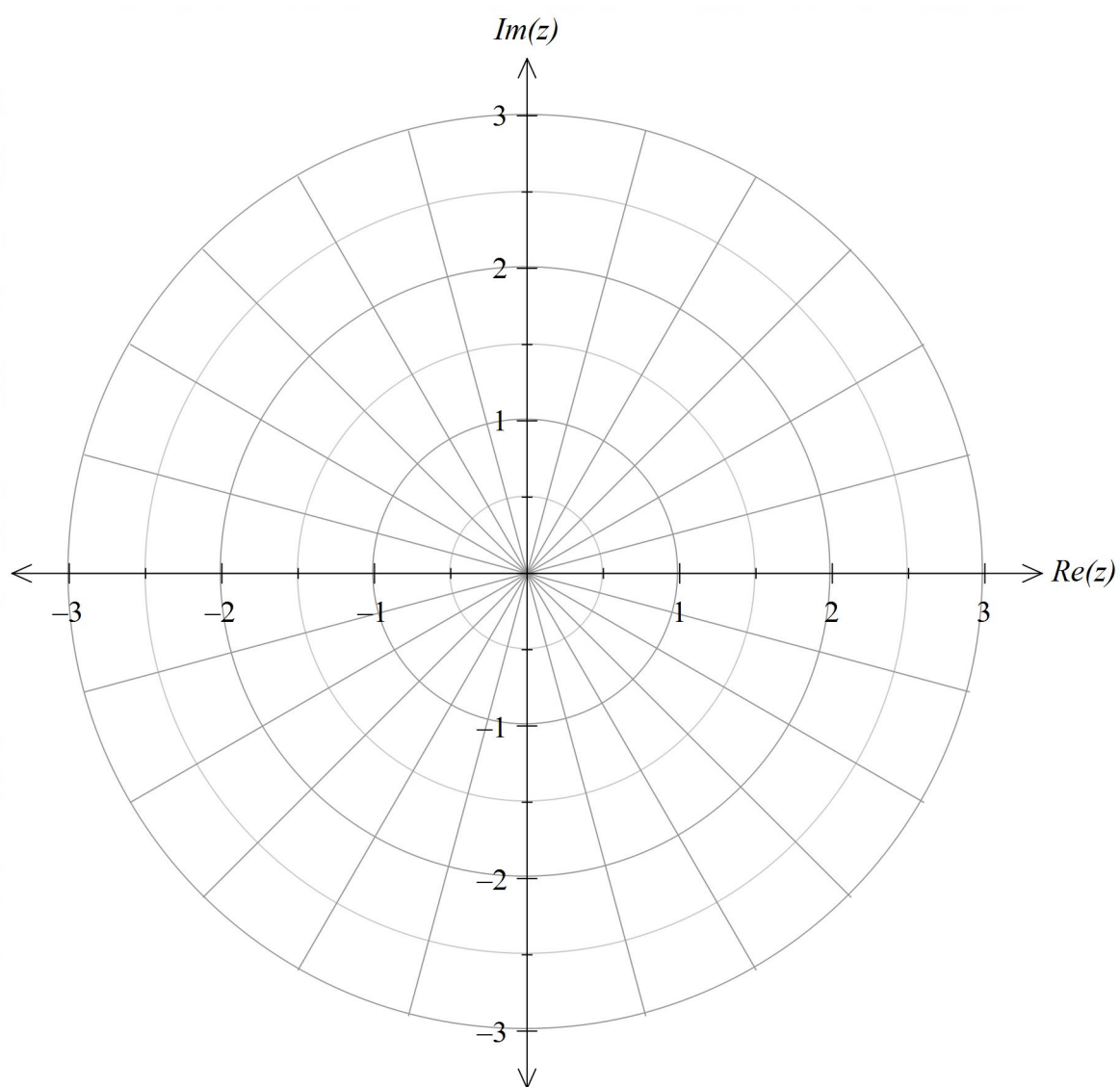
End of paper

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Argand diagram for Question 14(c).

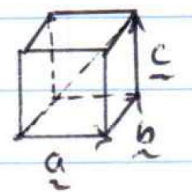
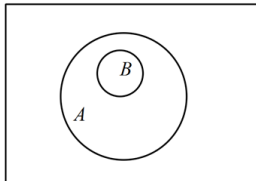
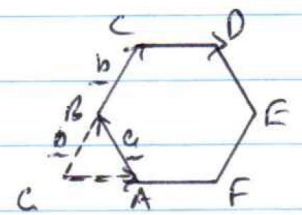


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Extension 2 Task 4 2023 Solutions

Section I

1.	$\text{Let } z = x + iy$ $z\bar{z} = (x+iy)(x-iy)$ $= x^2 + y^2$ $= z ^2$ $\bar{z}^{-1} = \frac{1}{x-iy} \times \frac{x+iy}{x+iy}$ $= \frac{x+iy}{x^2+y^2}$ $= \frac{z}{ z ^2}$ $\therefore$ both statements are true	C
2.	$\frac{4+2i}{1-2i} \times \frac{1+2i}{1+2i} = \frac{4+8i+2i-4}{1+4}$ $= \frac{10i}{5}$ $= 2i$ $\therefore \left \frac{4+2i}{1-2i} \right = 2$	D
3.	$A \Rightarrow B$ for contra positive: $\neg B \Rightarrow \neg A$	B
4.	$\int x^3 (x^4 - 1)^2 dx = \frac{1}{4} \int 4x^3 (x^4 - 1)^2 dx$ $= \frac{1}{4} \frac{(x^4 - 1)^3}{3} + C$ $= \frac{1}{12} (x^4 - 1)^3 + C$	D

5.	$v = k(a-x)$ $\frac{dx}{dt} = k(a-x)$ $\int_0^x \frac{dx}{a-x} = \int_0^t k dt$ $[-\ln a-x]_0^x = [kt]_0^t$ $-\ln a-x - -\ln a = kt$ $\ln\left \frac{a}{a-x}\right = +kt$ $\frac{a}{a-x} = e^{kt}$ $\frac{a-x}{a} = e^{-kt}$ $a-x = ae^{-kt}$ $x = a(1-e^{-kt})$	C
6.	G. $\hat{a} + \hat{b} + \hat{c}$ will be the diagonal of a unit cube.  $ \hat{a} + \hat{b} = \sqrt{1^2 + 1^2}$ $= \sqrt{2}$ $ \hat{a} + \hat{b} + \hat{c} = \sqrt{1^2 + 1^2 + 1^2}$ $= \sqrt{3}$	C
7.	If B implies A then Diagram below represents the $\neg A \Rightarrow \neg B$? 	B
8.	$\vec{CD} = \vec{CA}$ now $\vec{CA} = \vec{b} - \vec{a}$  $\therefore \vec{CD} = \vec{b} - \vec{a}$	C

9.

$$\int_{-\pi}^{\pi} x \sin x \, dx$$

$x \sin x$ is even and positive for $-\pi \leq x \leq \pi$
 $x \sin x$ attains a value of at least $\pi/2$ (when $x = \pi/2$)

$$\int_{-\pi/2}^{\pi/2} \cos(x^3) \, dx$$

$\cos(x^3)$ is even and attains a value of 1 (at $x = 0$); however, it is not positive for the entire domain $-\pi/2 \leq x \leq \pi/2$
 $\therefore < \int_{-\pi}^{\pi} x \sin x \, dx$.

$$\int_{-\pi}^{\pi} e^{-x^2} \, dx$$

e^{-x^2} is an even function and positive for $-\pi \leq x \leq \pi$
 e^{-x^2} attains a max value of 1 (at $x = 0$)

$$\therefore < \int_{-\pi}^{\pi} x \sin x \, dx.$$

$$\int_{-1}^1 \tan^{-1}(x^3) \, dx = 0$$

as $\tan^{-1}(x^3)$ is odd.

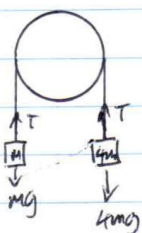
A

10.

$$F = 4mg - T - (mg - T) = 3mg.$$

$$\therefore (m+4m)a = 3mg.$$

$$a = \frac{3g}{5} \quad \therefore \text{not C}$$



$$\text{so } \frac{dv}{dt} = \frac{3g}{5}$$

$$v = \int_0^t \frac{3g}{5} \, dt$$

$$= \left[\frac{3gt}{5} \right]_0^t$$

$$= \frac{3gt}{5}$$

When $t = 4$.

$$v = \frac{3g \times 4}{5}$$

$$= \frac{12g}{5} \quad \therefore \text{not A.}$$

$$v = \frac{3gt}{5}$$

$$\frac{dx}{dt} = \frac{3gt}{5}$$

$$x = \int_0^t \frac{3gt}{5} \, dt$$

$$= \left[\frac{3gt^2}{10} \right]_0^t$$

$$= \frac{3gt^2}{10}$$

$$\text{at } t = 4 \quad x = \frac{3g}{10} \times 4^2$$

$$= \frac{48g}{10}$$

$$= \frac{24g}{5}$$

B

Section II

Question 11 (15 marks)

Marks

(a)	(i)	$z \bar{w} = (3+2i)(2+i) \quad \boxed{\checkmark}$ $= 6+3i+4i+2i^2$ $= 4+7i \quad \boxed{\checkmark}$	2
	(ii)	$\frac{z}{i w} = \frac{(3+2i)}{i(2+i)}$ $= \frac{3+2i}{2i-i^2}$ $= \frac{3+2i}{1+2i}$ $= \frac{3+2i}{1+2i} \times \frac{1-2i}{1-2i} \quad \boxed{\checkmark}$ $= \frac{3-6i+2i-4i^2}{1-4i^2}$ $= \frac{7-4i}{5} \quad \text{or} \quad \frac{7}{5} - \frac{4i}{5} \quad \boxed{\checkmark}$	2
(b)	$\int \frac{x+34}{(x-6)(x+2)} dx = \int \frac{A}{x-6} + \frac{B}{x+2} dx$ ie: $A(x+2) + B(x-6) = x+34$ $\left. \begin{array}{l} \text{When } x=6 \quad 8A+0=40 \\ \quad \quad \quad \therefore A=5 \\ \text{When } x=-2 \quad 0-8B=32 \\ \quad \quad \quad B=-4 \end{array} \right\}$ $\boxed{\checkmark}$ $\therefore \int \frac{x+34}{(x-6)(x+2)} dx = \int \frac{5}{x-6} + \frac{-4}{x+2} dx$ $= 5 \ln x-6 - 4 \ln x+2 + C$ $\boxed{\checkmark}$ $= \ln x-6 ^5 - 4 \ln x+2 + C$ $\boxed{\checkmark}$ $= \ln \frac{ x-6 ^5}{ x+2 } + C$ $\boxed{\checkmark}$		3

(c)	Prove that $\sqrt{3}$ is irrational. Proof by contradiction Assume $\sqrt{3}$ is rational $\left. \begin{aligned} \text{ie: } \sqrt{3} &= \frac{p}{q} \\ \text{where } p, q &\text{ are positive integers with no common factors other than 1} \end{aligned} \right\} \quad \boxed{\checkmark}$ So $\sqrt{3}q = p$ $3q^2 = p^2$ This implies that p^2 has a factor of 3. But square numbers have pairs of each factor by definition $\therefore p$ has a factor of 3 ie: $p = 3m$ So $3q^2 = (3m)^2$ $q^2 = 3m^2$ It follows that q^2 has a factor of 3 and so q has a factor of 3 So Both p and q have a factor of 3 which contradicts the assumption $\therefore \sqrt{3}$ is irrational by contradiction $\boxed{\checkmark}$	2
(d)	$\left. \begin{aligned} \int \frac{dx}{2x^2 + 3x + 4} &= \frac{1}{2} \int \frac{1}{x^2 + \frac{3}{2}x + 2} dx \\ &= \frac{1}{2} \int \frac{1}{x^2 + \frac{3}{2}x + \frac{9}{16} + 2 - \frac{9}{16}} dx \end{aligned} \right\} \quad \boxed{\checkmark}$ $= \frac{1}{2} \int \frac{1}{\left(x + \frac{3}{4}\right)^2 + \left(\frac{\sqrt{23}}{4}\right)^2} dx \quad \boxed{\checkmark}$ $= \frac{1}{2} \times \left(\frac{4}{\sqrt{23}} \tan^{-1} \left(\frac{\left(x + \frac{3}{4}\right)}{\frac{\sqrt{23}}{4}} \right) \right) + C$ $= \frac{2}{\sqrt{23}} \tan^{-1} \left(\frac{4x + 3}{\sqrt{23}} \right) + C \quad \boxed{\checkmark}$	3

(e)

$$P(x) = x^3 - x^2 - 7x + 15$$

Since the coefficients are rational, any complex roots occur in conjugate pairs (conjugate root theorem)

$\therefore$ if $\alpha = 2+i$ is a root then $\beta = 2-i$ is also a root.

$$\text{Now } \alpha + \beta = 2+i + 2-i \\ = 4$$

$$\alpha\beta = (2+i)(2-i) \\ = 5$$

$\therefore$ The quadratic with roots α & β is

$$x^2 - 4x + 5 = 0$$

$\therefore x^2 - 4x + 5$ is a factor of $P(x)$

$$\text{So } P(x) = (x^2 - 4x + 5)(x + 3)$$

By inspection

So roots: $2+i$, $2-i$, -3 .



OR

Sum of the roots

$$\alpha + \beta + \gamma = +1$$

$$(2+i)(2-i) + \gamma = 1$$

$$4 + \gamma = 1$$

$$\gamma = -3$$



3

Question 12 (15 marks)

Marks

(a)	Given $\int_2^3 f(u) du = \sqrt{7}$ $I = \int_1^2 \frac{1}{x^2} f\left(1 + \frac{x}{2}\right) dx.$ Let $u = 1 + \frac{x}{2}$ $du = \frac{1}{2} dx$ When $x=1$ $u = 1 + \frac{1}{2} = \frac{3}{2}$ When $x=2$ $u = 1 + \frac{2}{2} = 2$ $\therefore I = -\frac{1}{x} \int_1^2 -\frac{2}{x^2} f\left(1 + \frac{x}{2}\right) dx$ $= -\frac{1}{x} \int_{\frac{3}{2}}^2 f(u) du.$ $= \frac{1}{2} \int_2^3 f(u) du$ $= \frac{1}{2} (\sqrt{7})$ as u is dummy variable $= \frac{\sqrt{7}}{2}.$ OR $\int_2^3 f(x) dx = \sqrt{7}$ $\therefore [F(x)]_2^3 = \sqrt{7}$ $F(3) - F(2) = \sqrt{7}$ $I = \int_1^2 \frac{1}{x^2} f\left(1 + \frac{x}{2}\right) dx$ $= -\frac{1}{x} \int_1^2 -\frac{2}{x^2} f\left(1 + \frac{x}{2}\right) dx$ $= -\frac{1}{2} \left[F\left(1 + \frac{x}{2}\right) \right]_1^2$ as $\frac{d}{dx}\left(\frac{x}{2}\right) = \frac{1}{2}$ $= -\frac{1}{2} [F(2) - F(3)]$ $= \frac{1}{2} [F(3) - F(2)]$ $= \frac{\sqrt{7}}{2}$	3
(b)	(i) ii) Solving $x^3 - x^2 - 5x = 3 - y^2$ & $y = \sqrt{24}$ Simultaneously yields $x^3 - x^2 - 5x + 21 = 0$ This is a cubic so will have 3 roots. ☒ From the graph, there is only one point of intersection $\Rightarrow$ 1 real root, 2 complex roots. ☒	2

	(ii)	Sub $x = -3$ (from graph $x < 0$ & from eqⁿ x is odd) $P(x) = x^3 - x^2 - 5x + 21$ $P(-3) = (-3)^3 - (-3)^2 - 5(-3) + 21$ $= -27 - 9 + 15 + 21$ $= 0$ $\therefore x+3$ is a root of $P(x)$ ✓ i.e. $P(x) = (x+3)(x^2 - 4x + 7)$ by inspection The complex roots will be the solution to: $x^2 - 4x + 7 = 0$ $x^2 - 4x + 4 = -7 + 4$ $(x-2)^2 = -3$ $x-2 = \pm \sqrt{3}i$ $x = 2 \pm \sqrt{3}i$ $\therefore$ complex roots are ✓ $x = 2 + \sqrt{3}i, 2 - \sqrt{3}i$ → OR Let roots be α, β & -3 where $\alpha = x + iy, \beta = x - iy$ sum of roots $x + iy + x - iy + -3 = 1$ $2x = 4$ $x = 2$ Product of the roots $-3(x + iy)(x - iy) = -21$ $x^2 + y^2 = 7$ $(2)^2 + y^2 = 7$ $y^2 = 3$ $y = \pm \sqrt{3}$	2
(c)	(i)	$\vec{AB} = \vec{OB} - \vec{OA}$ $= 6\vec{b} - 3\vec{a}$ $= 3(2\vec{b} - \vec{a})$ ✓	1
	(ii)	$\vec{OY} = \vec{OB} + \vec{BY}$ $= 6\vec{b} + 5\vec{a} - \vec{b}$ $= 5(\vec{a} + \vec{b})$ ✓ Since $AX : XB = 1 : 2$ $\vec{AX} = \frac{1}{3} \vec{AB}$ Now $\vec{OX} = \vec{OA} + \vec{AX}$ $= 3\vec{a} + \frac{1}{3}(6\vec{b} - 3\vec{a})$ $= 3\vec{a} + 2\vec{b} - \vec{a}$ $= 2(\vec{a} + \vec{b})$ ✓ $\therefore \vec{OX} : \vec{OY} = 2(\vec{a} + \vec{b}) : 5(\vec{a} + \vec{b})$ $= 2 : 5$ so $\vec{OX} = \frac{2}{5} \vec{OY}$ ✓	3

(d)	(i)	We know, for $a, b \in \mathbb{R}$ $(a-b)^2 \geq 0$ $\therefore a^2 + b^2 - 2ab \geq 0$ $a^2 + b^2 \geq 2ab. \quad (1)$	✓	1
	(ii)	Similarly $b^2 + c^2 \geq 2bc \quad (2)$ $\& \quad c^2 + a^2 \geq 2ac \quad (3)$ Summing (1), (2) & (3): $a^2 + b^2 + b^2 + c^2 + c^2 + a^2 \geq 2ab + 2bc + 2ca$ $2a^2 + 2b^2 + 2c^2 \geq 2(ab + bc + ca)$ $a^2 + b^2 + c^2 \geq ab + bc + ca.$	✓	1
	(iii)	(iii) $(a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$ $\geq ab + bc + ca + 2(ab + bc + ca)$ from (ii) $= 3(ab + bc + ca)$ also $(a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$ $\leq a^2 + b^2 + c^2 + 2(a^2 + b^2 + c^2)$ from (ii) $= 3(a^2 + b^2 + c^2)$ $\therefore 3(ab + bc + ca) \leq (a+b+c)^2 \leq 3(a^2 + b^2 + c^2)$	✓	2

Question 13 (15 marks)

Marks

(a)	Step 1: Let $n=1$ Since there are no other disks it takes 1 move to transfer the disk from peg A to peg C $n=1$ moves $= 2^1 - 1$ $= 1$ $\therefore$ true for $n=1$ ✓ Step 2: Assume true for $n=k$ So it takes $2^k - 1$ moves to transfer the disks from peg A to peg C Step 3: Prove true for $n=k+1$ using the assumption in step 2, it will take $2^k - 1$ moves to transfer the top k disks from peg A to peg B it will then take 1 move to transfer the $(k+1)^{th}$ (last) disk from peg A to C it then takes $2^k - 1$ moves to transfer the k disks from peg B to peg C ✓ $\therefore$ moves $= (2^k - 1) + 1 + (2^k - 1)$ $= 2 \cdot 2^k - 1$ $= 2^{k+1} - 1$ as required. ✓ Step 4: The result holds by the inductive process.	3
(b)	(i) We have: $v \frac{dv}{dx} = 2x - 3x^2$ $\int_{2\sqrt{2}}^v v \frac{dv}{dx} dx = \int_0^x (2x - 3x^2) dx$ ✓ $\left[\frac{1}{2} v^2 \right]_{2\sqrt{2}}^v = \left[x^2 - x^3 \right]_0^x$ $\frac{1}{2} v^2 - \frac{1}{2} (2\sqrt{2})^2 = x^2 - x^3$ $v^2 - 8 = 2x^2 - 2x^3$ ✓ $v^2 = 8 + 2x^2 - 2x^3$	2

(ii)

Hence

$$v = \pm \sqrt{8 + 2x^2 - 2x^3}$$

But, when $x=0$, $v=2\sqrt{2}$

$$\therefore v = \sqrt{8 + 2x^2 - 2x^3}$$

For v to exist:

$$8 + 2x^2 - 2x^3 \geq 0$$

$$\text{or } x^3 - x^2 - 4 \leq 0$$

Let $x=2$:

$$2^3 - 2^2 - 4 = 0$$

$\therefore (x-2)$ is a factor of $x^3 - x^2 - 4$

So $(x-2)(x^2 + 2x + 2) \leq 0$ by inspection

So $x-2=0$ or $x^2 + 2x + 2=0$

$$x=2$$

$$\Delta = 1^2 - 4(1)(2)$$

$$< 0$$

$\therefore x=2$ is only real solution

$\therefore x \leq 2$ (as $x=0$ is initial condition)

2

(c)

Since O is the centre of the circle

$$|\vec{OA}| = |\vec{OB}| = |\vec{OC}|$$

radii of circle.

$$\text{also } \vec{OA} = -\vec{OB}$$

$$\text{now } \vec{AC} = \vec{OC} - \vec{OA}$$

$$= \vec{OC} + \vec{OB} \quad (\vec{OA} = -\vec{OB})$$

$$\& \vec{CB} = \vec{OB} - \vec{OC}$$

If $\angle ACB = 90^\circ$ then

$$\vec{AC} \cdot \vec{CB} = 0$$

$$\vec{AC} \cdot \vec{CB} = (\vec{OC} + \vec{OB})(\vec{OB} - \vec{OC})$$

$$= \vec{OC} \cdot \vec{OB} - \vec{OC} \cdot \vec{OC}$$

$$+ \vec{OB} \cdot \vec{OB} - \vec{OB} \cdot \vec{OC}$$

$$= -|\vec{OC}|^2 + |\vec{OB}|^2$$

$$= 0$$

$$\text{as } \vec{a} \cdot \vec{a} = |\vec{a}|^2$$

$$\& |\vec{OB}| = |\vec{OC}|$$

$$\therefore \angle ACB = 90^\circ$$

3

(d)



Since the particle is moving with SHM, we have.

$$v^2 = \omega^2 (a^2 - x^2)$$

Now, period = $\frac{2\pi}{\omega}$

$$\therefore \frac{2\pi}{\omega} = \frac{\pi}{3}$$

$$\omega = 6$$



so $v^2 = 36 (a^2 - x^2)$

When $x = 0.48$, $v = \pm 2.16$

$$\therefore (2.16)^2 = 36 (a^2 - (0.48)^2)$$

$$0.1296 = a^2 - 0.2304$$

$$a^2 = 0.36$$

$$a = \pm 0.6$$



Using $x = a \sin(\omega t)$ for simple harmonic motion about O:

$$x = 0.6 \sin 6t$$

Now, when $v = 2.88$

$$(2.88)^2 = 36 (0.6)^2 - x^2$$

$$0.2304 = 0.36 - x^2$$

$$x^2 = 0.1296$$

$$x = \pm 0.36$$

In SHM, speed is greatest in the mean position, therefore we need to find when the particle is between

$$-0.6 \leq x \leq 0.36 \quad \text{and} \quad 0.36 \leq x \leq 0.6$$



When $x = 0.36$

$$0.36 = 0.6 \sin 6t$$

$$\sin 6t = 0.6$$

$$6t = 0.6435$$

$$t = 0.1073$$



Since period is $\frac{\pi}{3}$, time to get to $x = 0.6$

is $t = \frac{\pi}{3} \div 4$

$$= 0.2618 \text{ sec}$$

$$\therefore t \text{ when } v < 2.88 = 4 \times (0.2618 - 0.1073)$$

$$= 4 \times 0.1545$$

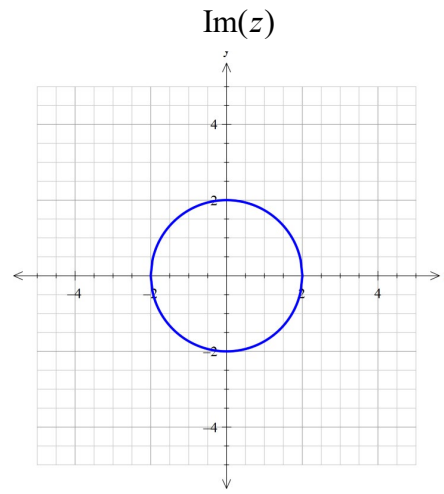
$$= 0.6182 \text{ sec}$$

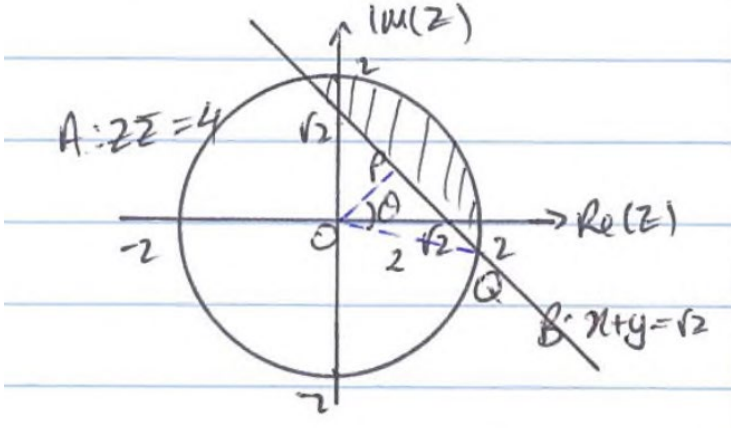


Question 14 (15 marks)

Marks

(a)	Prove $a^4 - 1$ is divisible by 16 for any odd 'a' Step 1: Let $a=1$ $1^4 - 1 = 0$ Which is true (but trivial) note if $a=3$ $3^4 - 1 = 81 - 1$ $= 80$ $= 16 \times 5$ ✓ $\therefore$ true for $a=3$ as well. Step 2: assume true for some odd number, k. ie $k^4 - 1 = 16M$ for some $M \in \mathbb{Z}$ OR Step 2: Assume true for some odd number $2k+1$ ie: $(2k+1)^2 - 1 = 16M$ where $M \in \mathbb{Z}$ Step 3: Prove true for next odd number $(2k+1)+2$ ie: Show $[(2k+1)+2]^4 - 1 = 16Q$ where $Q \in \mathbb{Z}$ $LHS = (2k+1)^4 + 4(2k+1)^3(2) + 6(2k+1)^2(2)^2 + 4(2k+1)(2)^3 + (2)^4 - 1$ $= (2k+1)^4 - 1 + 4(2k+1)^3(2) + 6(2k+1)^2(2)^2 + 4(2k+1)(2)^3 + (2)^4$ $= 16M + 4(2k+1)^3(2) + 6(2k+1)^2(2)^2 + 4(2k+1)(2)^3 + (2)^4$ by assumption $= 16M + 64k^3 + 192k^2 + 208k + 80$ $= 16(M + 4k^3 + 12k^2 + 13k + 5)$ $= 16Q$ where $Q \in \mathbb{Z}$	Step 3: Prove true for the next odd number, $a=k+2$ ie. $(k+2)^4 - 1 = 16N$, $N \in \mathbb{Z}$ $LHS = k^4 + 4(2)k^3 + 6(2)^2k^2 + 4(2)^3k + 2^4 - 1$ $= k^4 + 8k^3 + 24k^2 + 32k + 16 - 1$ $= 16M + 8k^3 + 24k^2 + 32k + 16$ from assumption $= 16(M + 2k + 1) + 8k^2(k+3)$ ✓ Now, since k is odd, $k+3$ is even. $\therefore k+3 = 2P$, $P \in \mathbb{Z}$ $= 16(M + 2k + 1) + 8k^2(2P)$ $= 16(M + 2k + 1 + k^2P)$ $= 16N$ as all terms in brackets are integers ✓ Step 4: The result holds by the inductive process.	3
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(b)	(i)	$I_n = \int_0^{\sqrt{3}} (3-x^2)^n dx$ (i) Integrating by parts with $u = (3-x^2)^n \quad v' = 1$ $u' = -2nx(3-x^2)^{n-1} \quad v = x$ ☒ $\therefore I_n = [x(3-x^2)^n]_0^{\sqrt{3}} - \int_0^{\sqrt{3}} -2nx^2(3-x^2)^{n-1} dx$ $= 0 + 2n \int_0^{\sqrt{3}} x^2(3-x^2)^{n-1} dx$ $= -2n \int_0^{\sqrt{3}} [(3-x^2)^{n-1}(3-x^2) - 3(3-x^2)^{n-1}] dx$ $= -2n \int_0^{\sqrt{3}} (3-x^2)^n dx + 6n \int_0^{\sqrt{3}} (3-x^2)^{n-1} dx$ $= -2n I_n + 6n I_{n-1}$ $I_n(2n+1) = 6n I_{n-1}$ $I_n = \frac{6n}{2n+1} I_{n-1}$ as required. ☒	3
	(ii)	$I_3 = \frac{6 \times 3}{2(3)+1} \cdot I_2$ $= \frac{18}{7} \cdot \frac{12}{5} \cdot 1$ $= \frac{18}{7} \times \frac{12}{5} \times \frac{6}{3} \int_0^{\sqrt{3}} dx$ $= \frac{432}{35} (\sqrt{3} - 0)$ $= \frac{432\sqrt{3}}{35}$ ☒	1
(c)	(i)	Let $z = x + iy$ Then $z\bar{z} = (x+iy)(x-iy)$ $= x^2 + y^2$ So $z\bar{z} = 4 \Rightarrow x^2 + y^2 = 4$ This is a circle, centre (0,0), radius 2 ☒ 	1

	(ii) (i) $z = z - 2cis\pi/4$ Now $2cis\pi/4 = 2\cos\pi/4 + 2i\sin\pi/4$ $= \sqrt{2}(1+i)$ then $z ^2 = z - \sqrt{2}(1+i) ^2$ ✓ $x^2 + y^2 = (x - \sqrt{2})^2 + (y - \sqrt{2})^2$ $x^2 + y^2 = x^2 - 2\sqrt{2}x + 2 + y^2 - 2\sqrt{2}y + 2$ $2\sqrt{2}(x+y) = 4$ $x+y = \frac{2}{\sqrt{2}}$ $x+y = \sqrt{2}$ x & y intercept is $\sqrt{2}$. ✓	2
	(iii)  ✓	1
	(iv) (iv) Distance OP on graph will be $\frac{1}{2}$ the distance from O to $2cis\pi/4$ as the line is the perpendicular bisector of O & $2cis\pi/4$ $2cis\pi/4 = \sqrt{2+2}$ $= 2$. ✓ $\therefore OP = 1$ also $OQ = 2$ (radius of circle) $\therefore \theta = \cos^{-1}(\frac{1}{2})$ $= \pi/3$ So Area of sector in circle of radius 2 subtended by an angle of $2(\pi/3)$ will be required area: $A = \frac{1}{2} 2^2 (\frac{2\pi}{3} - \sin\pi/3)$ $= \frac{4\pi}{3} - \sqrt{3}$ unit². ✓	2

(d)

d) Prove $\ln x \leq x-1$ for $x > 0$

consider $f(x) = x-1-\ln x$.

R.T.P. $f(x) \geq 0$ for $x > 0$

$$f'(x) = 1 - \frac{1}{x}$$

$f'(x) = 0 \Rightarrow$ max/min values

$$1 - \frac{1}{x} = 0$$

$$x = 1$$



$$f''(x) = \frac{1}{x^2} \Rightarrow f(x) \text{ concave up}$$

for all x so $x=1$

is absolute min of $f(x)$

$$f(1) = 1-1-\ln 1$$

$$= 0$$



$$\therefore f(x) \geq 0, x > 0$$

as required.

Question 15 (15 marks)

Marks

(a)	(i)	$r(\lambda) = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} + \lambda \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$ $A = (-1, 1, 2) \quad B = (1, 2, 4)$ $\vec{AB} = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} \quad P = (2, -3, 4)$ (i) $\vec{AP} = \begin{bmatrix} 2 - (-1) \\ -3 - 1 \\ 4 - 2 \end{bmatrix}$ $= \begin{bmatrix} 3 \\ -4 \\ 2 \end{bmatrix} \quad \checkmark$ $\text{Proj}_{\vec{AB}} \vec{AP} = \frac{\vec{b} \cdot \vec{AP}}{ \vec{b} ^2} \vec{b}$ $= \frac{2 \times 3 - 1 \times 4 + 2 \times 2}{2^2 + 1^2 + 2^2} \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$ $= \frac{6}{9} \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$ $= \frac{2}{3} \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} \quad \checkmark$	let d be perpendicular distance from P to l $\therefore d = \text{Proj}_{\vec{AB}} \vec{AP} - \vec{AP} $ $= \left \begin{bmatrix} \frac{4}{3} - 3 \\ \frac{2}{3} - (-4) \\ \frac{4}{3} - 2 \end{bmatrix} \right $ $= \sqrt{\left(-\frac{5}{3}\right)^2 + \left(\frac{14}{3}\right)^2 + \left(-\frac{2}{3}\right)^2}$ $= \sqrt{\frac{225}{9}}$ $= 5 \text{ units} \quad \checkmark$	3
	(ii)	let C be some point on the line l: $C = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} + \mu \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} \text{ for some } \mu.$ i.e. $C = (-1 + 2\mu, 1 + \mu, 2 + 2\mu)$ $\therefore \vec{AC} = \begin{bmatrix} 2\mu \\ \mu \\ 2\mu \end{bmatrix}$ If the area $\Delta ACP = 15 \cdot \mu^2$ we have $\frac{1}{2} \times \vec{AC} \times d = 15$ $\therefore \frac{1}{2} \times \vec{AC} \times 5 = 15$ $ \vec{AC} = 6 \quad \checkmark$		2

$$\text{So } \sqrt{(2\mu)^2 + (\mu)^2 + (2\mu)^2} = 6$$

$$9\mu^2 = 36$$

$$\mu^2 = 4$$

$$\therefore \mu = \pm 2$$

$$\text{So } C = (-1+4, 1+2, 2+4) \text{ if } \mu = 2$$

$$= (3, 3, 6)$$

$$\text{or } C = (-1-4, 1-2, 2-4) \text{ if } \mu = -2$$

$$= (-5, -1, -2)$$



(b) (i)

$$C = \cos \theta + \frac{1}{2} \cos 5\theta + \frac{1}{4} \cos 9\theta + \dots$$

$$S = \sin \theta + \frac{1}{2} \sin 5\theta + \frac{1}{4} \sin 9\theta + \dots$$

$$\text{ii) } C + iS$$

$$= (\cos \theta + i \sin \theta) + \frac{1}{2} (\cos 5\theta + i \sin 5\theta) + \frac{1}{4} (\cos 9\theta + i \sin 9\theta) + \dots$$

$$= \cos \theta + \frac{1}{2} i \sin \theta + \frac{1}{4} \cos 9\theta + \dots$$

$$= \cos \theta + \frac{1}{2} \cos \theta \cos 4\theta + \frac{1}{4} \cos \theta (\cos 4\theta)^2 + \dots$$

$$\text{Then is a G.P with } a = \cos \theta$$

$$r = \frac{1}{2} \cos 4\theta$$

So $C + iS$ will be the limiting sum of this series.



$$C + iS = \frac{\cos \theta}{1 - \frac{1}{2} \cos 4\theta}$$

$$= \frac{e^{i\theta}}{1 - \frac{1}{2} e^{4i\theta}}$$

$$= \frac{e^{i\theta}}{\frac{2 - e^{4i\theta}}{2}}$$

$$= \frac{2e^{i\theta}}{2 - e^{4i\theta}}$$

as required.



2

(ii)

To find S we need to rationalise the denominator of the expression in part (i)

$$C + iS = \frac{2e^{i\theta}}{2 - e^{4i\theta}} \times \frac{2 - e^{-4i\theta}}{2 - e^{-4i\theta}}$$

$$= \frac{4e^{i\theta} - 2e^{-3i\theta}}{4 - 2e^{-4i\theta} - 2e^{4i\theta} + e^0}$$



$$= \frac{4e^{i\theta} - 2e^{-3i\theta}}{4 - 2(e^{4i\theta} + e^{-4i\theta}) + 1}$$

$$\text{now } e^{4i\theta} + e^{-4i\theta}$$

$$= \cos 4\theta + \cos(-4\theta)$$

$$= \cos 4\theta + i \sin 4\theta + \cos(-4\theta) + i \sin(-4\theta)$$

$$= 2\cos 4\theta \text{ as } \sin(-4\theta) = -\sin 4\theta$$



$$\therefore C + iS = \frac{4e^{i\theta} - 2e^{-3i\theta}}{5 - 4\cos 4\theta}$$

$$S = \text{Im}(C + iS)$$

$$= \frac{4\sin \theta - 2\sin(-3\theta)}{5 - 4\cos 4\theta}$$

$$= \frac{4\sin \theta + 2\sin(3\theta)}{5 - 4\cos 4\theta}$$

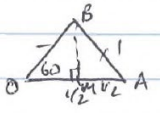


as required.

(c)	(i)	$A = \int_q^p (x-q)(x-p)^2 dx$ Integrate by parts: $\begin{aligned} \text{let } u &= x-q & v' &= (x-p)^2 \\ u' &= 1 & v &= \frac{1}{3}(x-p)^3 \end{aligned}$ $\therefore A = \left[\frac{1}{3}(x-q)(x-p)^3 \right]_q^p - \int_q^p \frac{1}{3}(x-p)^3 dx$ $= 0 + - \left[\frac{1}{12}(x-p)^4 \right]_q^p$ $= -\frac{1}{12}(0 - (q-p)^4)$ $= \frac{1}{12}(p-q)^4 \quad \text{as } (q-p)^4 = -(p-q)^4 = (p-q)^4$ ✓ ✓	2
	(ii)	To find the area of AMNB we need the y-value at c to give the height and know the base is $AB = p-q$. C is the turning point $\frac{dy}{dx} = (x-q) \cdot 2(x-p) + (x-p)^2 \cdot 1$ $= (x-p)(3x-2q-p)$ for max/min, $\frac{dy}{dx} = 0$ $\therefore (x-p)(3x-2q-p) = 0$ $x = p \text{ or } x = \frac{p+2q}{3}$ ✓ We know the turning point at b is when $x=p$ $\therefore$ at C, $x = \frac{p+2q}{3}$ $\begin{aligned} y &= \left(\frac{p+2q}{3} - q\right) \left(\frac{p+2q}{3} - p\right)^2 \\ &= \left(\frac{p+2q-3q}{3}\right) \left(\frac{p+2q-3p}{3}\right)^2 \\ &= \left(\frac{p-q}{3}\right) \left(\frac{2q-2p}{3}\right)^2 \\ &= \left(\frac{p-q}{3}\right) \left(\frac{-2(p-q)}{3}\right)^2 \\ &= \frac{4}{27} (p-q)^3 \end{aligned}$ ✓ $\therefore A = (p-q) \left(\frac{4}{27} (p-q)^3\right)$ $= \frac{4}{27} (p-q)^4$ $A_c: A_{AMNB} = \frac{1}{12} (p-q)^4 : \frac{4}{27} (p-q)^4$ $= \frac{1}{12} : \frac{4}{27}$ $= \frac{1}{4} : \frac{4}{9}$ $= 1 : 16/9$ ✓	3

Question 16 (15 marks)

Marks

(a)	a) Let C have coordinates (x, y, z) Given $\vec{OA} = \vec{OC} = 1$ ΔAOC is equilateral $\Rightarrow \angle AOC = \pi/3$ $\therefore \vec{OA} \cdot \vec{OC} = \vec{OA} \vec{OC} \cos \pi/3$ i.e. $(\frac{1}{2}) \cdot (\frac{1}{2}) = 1 \times 1 \times \frac{1}{2}$ $x + 0 + 0 = \frac{1}{2}$ $x = \frac{1}{2}$ ✓ Now B is the altitude of ΔOAB.  $\sin 60^\circ = \frac{OB}{OA}$ $= \frac{\sqrt{3}}{2}$ $\therefore B = (\frac{1}{2}, \frac{\sqrt{3}}{2}, 0)$ ✓ So $\vec{OB} \cdot \vec{OC} = \vec{OB} \vec{OC} \cos \pi/3$ $(\frac{1}{2}, \frac{\sqrt{3}}{2}, 0) \cdot (\frac{1}{2}, \frac{1}{2}, z) = 1 \times 1 \times \frac{1}{2}$ $\frac{1}{4} + \frac{\sqrt{3}}{2}y + 0 = \frac{1}{2}$ $\frac{\sqrt{3}}{2}y = \frac{1}{4}$ $y = \frac{1}{2\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$ $= \frac{\sqrt{3}}{6}$ ✓	Finally $\vec{OC} = 1$ $\therefore \sqrt{(\frac{1}{2})^2 + (\frac{\sqrt{3}}{6})^2 + z^2} = 1$ $\frac{1}{4} + \frac{3}{36} + z^2 = 1$ $z^2 = 1 - \frac{1}{4} - \frac{1}{12}$ $= \frac{2}{3}$ $z = \sqrt{\frac{2}{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$ $= \frac{\sqrt{6}}{3}$ ✓ $\therefore C = (\frac{1}{2}, \frac{\sqrt{3}}{6}, \frac{\sqrt{6}}{3})$	4
(b)	(i) for $a, b \in \mathbb{R}^+$ we have — $(\sqrt{\frac{a}{b}} - \sqrt{\frac{b}{a}})^2 \geq 0$ $\frac{a}{b} - 2\sqrt{\frac{a}{b} \cdot \frac{b}{a}} + \frac{b}{a} \geq 0$ $\frac{a}{b} - 2 + \frac{b}{a} \geq 0$ $\frac{a}{b} + \frac{b}{a} \geq 2$ ✓ as required.	OR $(a-b)^2 \geq 0$ $a^2 + b^2 \geq 2ab$ $\frac{a^2 + b^2}{ab} \geq 2$ where $(ab \geq 0)$ $\frac{a^2}{ab} + \frac{b^2}{ab} \geq 2$ $\frac{a}{b} + \frac{b}{a} \geq 2$	1
(ii)	Prove $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}$ From (i) we have: $\frac{a+b}{b+c} + \frac{b+c}{a+b} \geq 2$ (*) ✓ So $\frac{a+b}{b+c} + \frac{b+c}{c+a} + \frac{c+a}{c+a} + \frac{c+a}{a+b} + \frac{a+b}{a+b} + \frac{a+b}{c+a}$ $\geq 2 + 2 + 2$ ✓	from summing the 3 variations of (*) Rearranging: $\frac{a+b}{b+c} + \frac{c+a}{b+c} + \frac{b+c}{c+a} + \frac{a+b}{c+a} + \frac{b+c}{a+b} + \frac{c+a}{a+b} \geq 6$ $\frac{2a}{b+c} + \frac{b+c}{b+c} + \frac{2b}{c+a} + \frac{c+a}{c+a} + \frac{2c}{a+b} + \frac{a+b}{a+b} \geq 6$ $2(\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b}) + 3 \geq 6$ $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}$ as required.	3

(c)	(i)	$F = mg - kv$ $\therefore ma = mg - kv$ or $a = g - kv$ since $m = 1 \text{ kg}$. $\therefore \frac{dv}{dt} = g - kv \quad \checkmark$ Integrating both sides w.r.t. time: $\int_0^v \frac{dv}{g - kv} = \int_0^t dt$ $\left[-\frac{1}{k} \ln g - kv \right]_0^v = [t]_0^t$ $-\frac{1}{k} (\ln(g - kv) - \ln g) = t$ $\ln \frac{g - kv}{g} = -kt$ $\frac{g - kv}{g} = e^{-kt}$ $g - kv = g e^{-kt}$ $kv = g - g e^{-kt}$ $v = \frac{g}{k} (1 - e^{-kt}) \quad \checkmark$ as required.	2
	(ii)	Again, we have $a = g - kv$ or $\frac{dv}{dt} = g - kv$ Integrating w.r.t time with adjusted limits we have: $\int_u^v \frac{dv}{g - kv} = \int_2^t dt \quad \checkmark$ $\left[-\frac{1}{k} \ln g - kv \right]_u^v = [t]_2^t$ $-\frac{1}{k} (\ln(g - kv) - \ln(g - ku)) = t - 2$ $\ln \left(\frac{g - kv}{g - ku} \right) = -k(t - 2)$ $\frac{g - kv}{g - ku} = e^{-k(t - 2)}$ $g - kv = (g - ku) e^{-k(t - 2)}$ $kv = g - (g - ku) e^{-k(t - 2)}$ $v = \frac{g}{k} - e^{-k(t - 2)} \left(\frac{g - ku}{k} \right) \quad \checkmark$ as required.	2

(iii)

Sub $u = g/k$ in the expression for V in (i)

$$V = \frac{g}{k} - e^{-k(t-2)} \left(\frac{g - k(g/k)}{k} \right)$$

$$= \frac{g}{k} - e^{-k(t-2)} \left(\frac{g - g}{k} \right)$$

$$= \frac{g}{k}$$



$$\therefore \frac{dy_2}{dt} = \frac{g}{k}$$

$$\int_0^{y_2} dy = \int_2^t \frac{g}{k} dt$$

$$y_2 = \frac{g}{k}(t-2)$$



2

(iv)

for collision, $y_1 = y_2$

$$\therefore \frac{g}{k} \left(t + \frac{1}{k}(e^{-kt} - 1) \right) = \frac{g}{k}(t-2)$$

$$t + \frac{1}{k}(e^{-kt} - 1) = t-2$$

$$e^{-kt} - 1 = -2k$$

$$e^{-kt} = 1-2k$$

$$-kt = \ln |1-2k|$$

$$t = -\frac{1}{k} \ln |1-2k|$$

as required.



1